

THE HAND OF THE MUSICIAN: THE KINEMATICS OF THE BIDIGITAL FINGER SYSTEM WITH ANATOMICAL RESTRICTIONS

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Abstract—Tendons of the fingers are frequently interconnected by anatomic structures that limit the displacements of these tendons relative to each other. In this paper a bidigital finger system in which such interconnections between tendons are present is kinematically modelled. Using this model, an exhaustive description of the effects on finger movement of connections between the different tendons of the fingers is given. The study provides a context for the interpretation of typical difficulties of finger coordination in musicians, especially in pianists and string players.

NOMENCLATURE

MCP	metacarpophalangeal joint
PIP	proximal interphalangeal joint
DIP	distal interphalangeal joint
FP	flexor digitorum profundus
FS	flexor digitorum superficialis
ED	extensor digitorum
IO	interosseus
MB	medial band of extensor and interosseus
LB	lateral band of extensor and interosseus
M_e	common equivalent motor
$T_c^{(i)}$	connected tendon of finger F_i
r_{ij}	moment arms of the tendons over the respective joints.
The first index denotes the motor: P=FP, S=FS, E=ED, I=IO, M=MB, L=LB. The second index numbers the joint: MCP=1, PIP=2, DIP=3 (see Table 1)	
$\delta_T^{(i)}$	displacement of a tendon T_i
$\theta_j^{(i)}$	joint rotations of finger F_i ($j=1, 2, 3$ for MCP, PIP, DIP, respectively)
$\sigma_T^{(i)}$	slack in tendon T of finger F_i
σ_c	slack in the connection between tendons
L_c	length of the connection between tendons
T_p	tendon part proximal to the insertion of the connection
T_d	tendon part distal to the insertion of the connection
L_i	distance between the fixed reference point and the insertion of the connection in tendon T_i

INTRODUCTION

Tendons of the fingers in the human hand are frequently interconnected by tendinous or fascia-like anatomical structures. Well-known examples are the juncturae tendinum between the M. extensor

digitorum (Forbes, 1991; Kaplan, 1965; Von Schroeder *et al.*, 1990), and connections within the M. flexor digitorum profundus (the phenomenon of the quadriga of Verdan) (Verdan, 1960; Fahrner, 1971; Malerich *et al.*, 1987), but many other instances have been described (Austin *et al.*, 1989; Baker *et al.*, 1981; Blair and Omer, 1981; Culver, 1980; Linburg and Comstock, 1979; McGregor and Glover, 1988; Parrot and Harrison, 1980). When sufficiently stiff, these structures limit the mutual displacements of the tendons they connect, and can be modelled as strict kinematic constraints. In this paper the effects of such constraints on the kinematics of finger movement are studied in a two finger model. To this end, the bidigital model with tendons connected with each other is required to execute some simple tasks at the piano keyboard, such as the lifting of fingers, or the striking of keys. The various ways in which the finger model can execute these tasks are calculated, and the effects of the constraints are pointed out. The connections modelled are inextensible, and between tendons of (i) the extensor digitorum (ED), (ii) the flexor digitorum superficialis (FS) and (iii) the flexor digitorum profundus (FP).

In the discussion, mention is made of extensible anatomical constraints, as may exist between muscles. Their effect on movement is compared with the inextensible connections studied in the model, and the findings are discussed with respect to the surgical clearance of the connections. Also discussed is the exerciseability of the hand with anatomic constraints, and the relative applicability of pedagogic models in the instrumental technique.

The present paper is a direct sequel to the paper of Leijnse *et al.* (1992), in which the kinematic functioning of a monodigital system with anatomical constraints was modelled.

METHODS AND MATERIALS

The kinematic two finger model with inextensible connections between tendons (Fig. 1)

The two-finger model studied consists of two equal sized, two-dimensional digits moving in parallel planes perpendicular to the joint axes. The functioning of the individual finger is described by the model of Leijnse *et al.* (1992). This model is based on the simplified representation of the finger as shown in Fig. 1., and specifically describes the controllable movements feasible in the free-moving finger (i.e. the movements that can be realised and controlled by the mere pulling of the tendons). The equations are:

$$d\epsilon_{FP} = r_{P1} d\theta_1 + r_{P2} d\theta_2 + r_{P3} d\theta_3 - d\sigma_P, \quad (1a)$$

$$d\epsilon_{FS} = r_{S1} d\theta_1 + r_{S2} d\theta_2 - d\sigma_S, \quad (1b)$$

$$d\epsilon_{ED} = -r_{E1} d\theta_1 - r_{M2} d\theta_2, \quad (1c)$$

$$d\epsilon_{IO} = r_{I1} d\theta_1 - r_{M2} d\theta_2, \quad (1d)$$

$$d\theta_3 = \frac{r_{M2} - r_{L2}}{r_{L3}} d\theta_2 - \frac{d\sigma_L}{r_{L3}}, \quad (1e)$$

$$\sigma_P * \sigma_S = 0. \quad (1f)$$

The motors are the flexor digitorum profundus (FP), the flexor digitorum superficialis (FS), the extensor digitorum (ED), the interosseus (IO). The Lumbricalis is ignored. The ED and IO insert in both the medial and the lateral bands (MB, LB) of the extensor assembly, which inserts in the basis of the middle (MB) and end phalanx (LB). In this kinematic model, the motors are instances that generate, or take up tendon displacement. The parameters of the model are the lengths of the moment arms of the motors over the respective joints (r_{ij}), and the lengths of the phalanges as measured between successive joint axes. The values used for calculation are given in Table 1. The moment

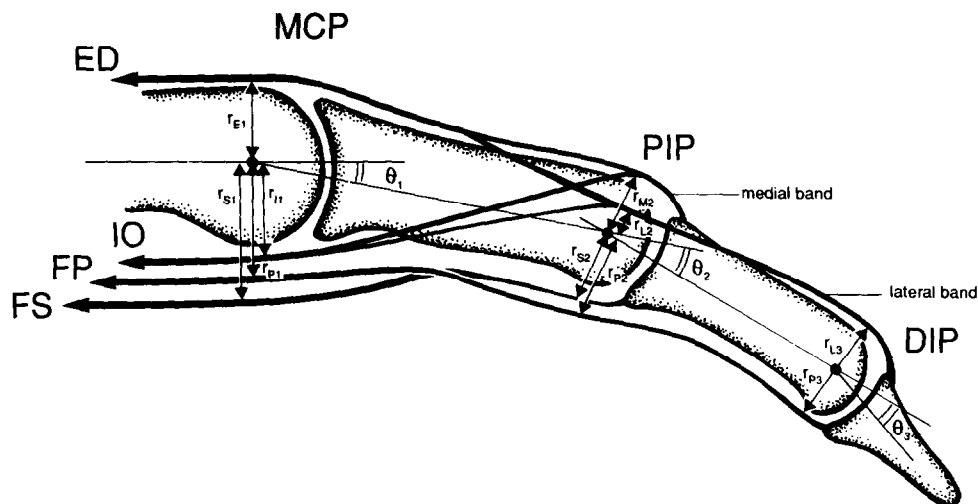


Fig. 1. Model of the finger.

Table 1. Moment arms of the tendons.

Motor joint	Symbol†			Value‡		
	MCP	PIP	DIP	MCP	PIP	DIP
FP	r_{P1}	r_{P2}	r_{P3}	11	10.5	6
FS	r_{S1}	r_{S2}	/	13	8.5	/
ED _{MB}	r_{E1}	r_{M2}	/	9	5	/
IO _{MB}	r_{I1}	r_{M2}	/	6	5	/
ED _{LB}	r_{E1}	r_{L2}	r_{L3}	9	*	4
IO _{LB}	r_{I1}	r_{L2}	r_{L3}	6	*	4

$$* \quad r_{L2} = r_{M2} - r_{L3} \left(0.25 + \frac{0.55}{\pi/2} \theta_2 \right)$$

Proximal, Middle, Distal phalanx: 50, 34, 17

†Notations used in the text.

‡Real values (in mm) used to calculate the results [From: Spoor (1983)].

/ indicates that the tendon does not cross the joint.

arms for extension have a negative sign. The θ_j are the joint rotations, while the ε_M are the changes of length of the motors as measured between their origin and insertion. Motors are allowed to be active and inactive; in an inactive motor the tendon is slack by definition, and in an unconstrained active motor it is taut. The tendon of an active motor can only be slack when its free course is obstructed, e.g. by an anatomical connection, and the slack can only occur in the part of the tendon distal to the obstruction. The slack in a tendon is quantified by the slack variables σ_T . In the model (1), the slack variables are only indicated for tendons that may be slack without disrupting normal finger function in the unloaded finger: FS, FP and LB. Hereby the condition (1f) (at least one flexor must be taut) must be satisfied.

Anatomical connections are modelled as inextensible ropes of length L_c , connecting two tendons, one in each finger ($T_c^{(1)}$ and $T_c^{(2)}$ in F_1 and F_2 , respectively; c stands for connected), and inserting in the tendons proximal to the MCP (see Fig. 2). Only connections between tendons of the same muscle group will be considered, e.g. ED⁽¹⁾ to ED⁽²⁾ and FS⁽¹⁾ to FS⁽²⁾, etc. The tendons themselves run parallel and at zero distance from each other, so that the connection does not pull them laterally. In the connected tendon, the tendon part proximal to the insertion of the connection is named $T_p^{(i)}$, the part distal to it $T_d^{(i)}$. The tendon lengths between the reference point and the points of insertion of the connection are L_1 and L_2 for $T_c^{(1)}$ and $T_c^{(2)}$, respectively.

Displacement properties of connected tendons

In the following the displacement properties of the system with connected tendons are summarised.

(1) Connected tendons move unconstrained as long as the connection is slack. Let $\Delta\varepsilon_{T_c}$ be defined as

$$\Delta\varepsilon_{T_c} = \varepsilon_{T_c}^{(1)} - \varepsilon_{T_c}^{(2)} \quad (2)$$

and let in the initial position of a movement

$$\Delta\varepsilon_{0T_c} = 0,$$

$$\Delta L_0 = L_{01} - L_{02} \quad (\text{with } |\Delta L_0| \leq L_c). \quad (3)$$

From this initial position the maximal relative displacement possible between the connected tendons is

$$-(L_c - \Delta L_0) \leq \Delta\varepsilon_{T_c} \leq L_c + \Delta L_0. \quad (4)$$

The range of independent tendon displacements in a given finger position is thus determined by two parameters: (i) the length L_c of the connection; and (ii) the relative position ΔL_0 of the insertions of the

connection. Expression (4) can be rewritten as

$$\sigma_c = L_c - |\Delta\varepsilon_{T_c} - \Delta L_0| \geq 0 \quad (5)$$

in which σ_c is the slack in the connection, defined as the (positive) displacement needed to taut the connection.

(2) When taut, the connection connects a proximal part of one tendon, with a distal part of another. This will be used to indicate the orientation of the connection, e.g. $T_p^{(1)}T_d^{(2)}$ would mean that the connection is stretched between the proximal part $T_p^{(1)}$ of $T^{(1)}$ and the distal part $T_d^{(2)}$ of $T^{(2)}$. Tautness of the connection implies the tautness of the proximal and distal tendon parts it connects. The other tendon parts may be either taut or slack. In the slack notation

$$\sigma_c = 0 \Rightarrow \begin{cases} \sigma_{T_p}^{(1)} + \sigma_{T_d}^{(2)} = 0 & \text{and } \sigma_{T_d}^{(1)}, \sigma_{T_p}^{(2)} \geq 0, \\ \text{or} \\ \sigma_{T_d}^{(1)} + \sigma_{T_p}^{(2)} = 0 & \text{and } \sigma_{T_p}^{(1)}, \sigma_{T_d}^{(2)} \geq 0. \end{cases} \quad (6)$$

(3) Most generally, movements of fingers with connected tendons are partly constrained [the phases of the movement during which the connection is taut ($\sigma_c = 0$)], and partly unconstrained [the phases in which the connection is slack ($\sigma_c \neq 0$)]. When the connection becomes taut and remains taut during a movement it holds that $\sigma_c = d\sigma_c = 0$. Substitution of equation (5) into the latter equality leads to

$$dL_c = \text{sign} [\Delta\varepsilon_{T_c} - \Delta L_0]_{\sigma_c=0} \cdot d\Delta\varepsilon_{T_c} \quad (7)$$

From this follows that for taut and inextensible connections ($dL_c = 0$), the displacements of the connected tendons must be equal:

$$\left. \begin{array}{l} \sigma_c = 0 \\ d\sigma_c = 0 \\ dL_c = 0 \end{array} \right\} \Rightarrow d\Delta\varepsilon_{T_c} = 0 \Rightarrow d\varepsilon_{T_c}^{(1)} = d\varepsilon_{T_c}^{(2)}. \quad (8)$$

Inextensible connections of zero length

In this paper, only (phases of) constrained movements, i.e. movements for which $\sigma_c = 0$, will be investigated. Hereby L_c will first be put to zero without loss of generality, since in condition (8) the length L_c is not a parameter. This allows to avoid tedious descriptions of the initial state of the system (definitions of L_{0i} , orientation of the connection, etc.). Physically, tendons connected with connections of zero length ($L_c = 0$) can be imagined as being firmly sewn together proximal to the MCP. In the examples it is assumed that in the tendons thus connected the lengths of the tendon parts distal to the connection are equal, so that when both fingers are in identical positions ($\theta_j^{(1)} = \theta_j^{(2)}$, $j = 1, 2, 3$) these tendon parts are equally taut or slack.

The common equivalent motor

Condition (8) states that when the connection is taut all connected motors have the same displace-

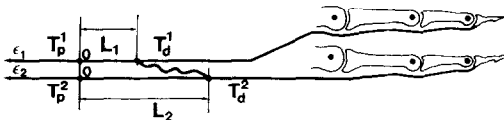


Fig. 2. Model of the bidigital system with connections between tendons.

ment. This means that when a tendon is displaced proximally (by the contraction of its motor), all *tendons* connected to it must follow this proximal displacement (otherwise this displacement is not possible); conversally, when a tendon is displaced distally (by pulling its distal end), all *motors* connected to it must elongate to follow this distal displacement, otherwise they would disallow it. Therefore, the motors of the tautly connected tendons can be replaced, without changing the functioning of the system, by a single equivalent motor, which for further use will be called the *common equivalent motor* M_e (Fig. 3). For the common motor it holds that

$$d\epsilon_{M_e} = d\epsilon_{T_c}^{(1)} = d\epsilon_{T_c}^{(2)} \quad (9)$$

with

$$d\epsilon_{T_c}^{(i)} = \sum_{j=1}^{n_{T_c}} \pm r_{T_j}^{(i)} d\theta_j^{(i)} - \sigma_{T_c}^{(i)}, \quad (10)$$

where $T_c^{(i)}$ is the coupled tendon of finger F_i , and n_{T_c} the number of joints this tendon crosses.

In analogy with the unconnected motor, the common motor may be called active when at least one of the connected tendons is taut, and inactive when all connected tendons are slack. In the slack notation:

$$M_e \text{ active} \Leftrightarrow \sigma_{T_c}^{(1)} * \sigma_{T_c}^{(2)} = 0. \quad (11)$$

Elementary bidigital tasks: notations

In this section notations for the bidigital tasks are defined. For further use such tasks are called 'functional problems', and any movement that is able to execute the task (solve the problem) is called a 'solution'. An elementary bidigital problem $p = (p_1, p_2)$ consists of the elementary monodigital problems p_1 and p_2 for the individual fingers F_1 and F_2 . A solution $s = (s_1, s_2)$ to a bidigital problem p consists of the solutions s_i of the individual problems p_i for the respective fingers F_i .

Only two elementary monodigital problems are distinguished here:

- (1) 'strike a key' [notation: 'k' (key-stroke)],
- (2) 'lift the finger tip from the key to a sufficient height ($h_{tip} > H$)' [notation: 'l' (lift)].

Four elementary bidigital problems are thus

possible:

$$\{(l, l), (k, l), (l, k), (k, k)\}.$$

The problem is completely specified when the initial finger positions $i = (i_1, i_2)$ are given. The initial positions i_j of the single digit can be partitioned into two classes C_1 and C_2 , as follows:

C_1 : all positions of holding down a key [notation: D (down)],

C_2 : all positions of keeping the finger lifted at a sufficient height $h_{tip} > H$ [notation: U (up)].

This leads to four classes C_i of bidigital initial positions:

$$\{(D, D), (D, U), (U, D), (U, U)\}.$$

A completely specified functional problem p can now be represented as

$$p = [(i_1, i_2), (p_1, p_2)].$$

In the model, the solving of a problem p is the production of the appropriate end positions $e = (e_1, e_2)$. Thereby the classes of the possible end positions C_e are equal to the classes of the possible initial positions C_i , as any end position may serve as the initial position of a subsequent functional problem.

Monodigital problems p_j in which the required end position e_j is in the same class C_e as the begin position i_j ($C_e = C_i$) will be called *identity problems*. Solving the identity problem is, by definition, the maintaining of the functional position of the finger, i.e. keeping the finger tip at a sufficient height, or holding the key down. As will be further illustrated, this does not mean that the finger must remain immobile, since $C_e = C_i$ does not imply that the specific end position e_j must be equal to the specific begin position i_j .

Example: $p = [(D, D), (l, k)]$

Interpretation:

(i) F_1 initially holds down a key ($i_1 = D$), and has to lift to a sufficient height ($p_1 = l$). F_1 will solve the problem by changing its position from the initial position 'D' to the end position 'U'.

(ii) F_2 holds down a key ($i_2 = D$), and should strike this key ($p_2 = k$). This is an identity problem, since F_2 already holds down the key.

The total solution will be that F_1 lifts to a sufficient height, while F_2 keeps holding its key down. Hereby F_2 may change its position ($e_2 \neq i_2$ or $\theta_{ji}^{(2)} \neq \theta_{je}^{(2)}$, $j = 1 \dots 3$), as long as its fingertip remains on the key ($h_{Fip}^{(2)} = h_{key}$).

The presentation of the data

In the following the finger movements are calculated from the parameter values of Table 1. They are presented with stick diagrams, by their begin and end positions. The excursions (excursion is a clinical term indicating 'displacement') of the motors required to bring the finger from the begin position to the end position are presented with bars in the stick diagram of the end position (Fig. 5).

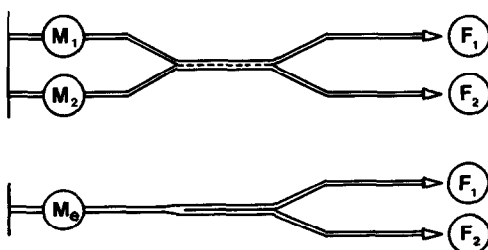


Fig. 3. The common equivalent motor M_e .

RESULTS

The bidigital system with strictly connected extensor tendons (ED) (Fig. 4)

The coupling of the displacements of the finger joints as resulting from a taut and strict ($L_c=0$) connection is quantified. From equation (9) follows (with $M_e=ED_e$):

$$d\epsilon_{ED_e} \stackrel{(a)}{=} d\epsilon_{ED}^{(1)} \stackrel{(b)}{=} d\epsilon_{ED}^{(2)}. \quad (12)$$

This condition (12) expresses that in the strictly connected system only movements are feasible in which the connected tendons have the same displacement (excursion). For further use, such movements will be called *iso-excursion movements*, bearing in mind that the iso-excursion applies to the connected tendons only. Substitution of (1.c) in (12.b) gives the resulting coupling in finger movement:

$$r_{E1} d\theta_1^{(1)} + r_{M2} d\theta_2^{(1)} = r_{E1} d\theta_1^{(2)} + r_{M2} d\theta_2^{(2)}. \quad (13)$$

In the following five different problems will be investigated, each illustrating different aspects of the iso-excursion movements.

Problem [(U, U), (k, k)] [Fig. 5(a)]. Both fingers are initially lifted in an identical position ($i_1=i_2$ or $\theta_{0j}^{(1)}=\theta_{0j}^{(2)}$, $j=1 \dots 3$), and they have to perform the identical tasks of lifting the finger ($p_1=p_2=k$). Such problems, i.e. problems in which the initial positions and the tasks are identical for both fingers, will be further called *unimodal problems*.

Solution. The obvious way to solve an unimodal problem ($p_1=p_2$) is by an *unimodal solution* ($s_1=s_2$). Since then holds that $d\theta_j^{(1)}=d\theta_j^{(2)}$ ($j=1 \dots 3$), condition (13) [or most generally condition (9)] is satisfied whatever the joint rotations $d\theta_j$. The physical interpretation is that when both fingers move in an identical way, no relative displacement of the connected tendons occurs (the fingers being of equal dimension, which we assumed), and the connection merely displaces together with the tendons without causing any restriction of movement.

Problem [(U, U), (k, l)] [Fig. 5(b)]. Initially, both fingers are lifted with all joints extended ($\theta_{0j}^{(i)}=0$,

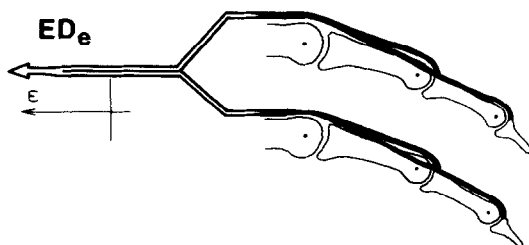


Fig. 4. Model of the bidigital system with connected extensors.

$j=1 \dots 3, i=1, 2$). From this position, F_1 has to strike a key, while F_2 has to solve the identity problem (i.e. maintain its lifting height $h_{tip}^{(2)} \geq H$).

Solution. In order to be able to strike the key, F_1 must flex in at least one joint. To allow this, the common motor ED_e must displace distally, say by a quantity: ϵ_{ED_e} . Hereby the extensor tendon $ED_e^{(2)}$ of the coupled finger F_2 will follow the displacement of the common motor ED_e , while remaining taut as expressed by condition (1c). This poses the problem of how to distribute the distal displacement ϵ_{ED_e} of the common motor over the joints of finger F_2 , while maintaining its lifting height ($h_{tip}^{(2)} \geq H$). Within the model, the problem is formulated as follows: from the initial positions (i_1, i_2), produce the end positions (e_1, e_2), such that:

$$h_{Fup}^{(2)} - h_{Fup}^{(1)} \geq H \quad (14)$$

(hereby is $h_{Fup}^{(1)}=h_{key}$), while satisfying condition (13). The problem is solved in the appendix. It is found that an infinity of feasible solutions exists, one of which is presented in Fig. 5(b). However, characteristic for all solutions is that the lifted finger F_2 shifts into a collapse position (Leijnse *et al.*, 1992), i.e. an intrinsic-minus-like position with a hyperextended MCP and a severely flexed PIP. The reason for this can be physically understood. The distal displacement ϵ_{ED_e} of tendon $ED_e^{(2)}$ due to the distal displacement of the ED_e will cause flexion in either or both the MCP and PIP of F_2 . The loss of height of the finger tip is greater when the MCP flexes, than when the PIP flexes. Therefore, to minimise the loss of height, the displacement ϵ_{ED_e} must be taken up entirely by the PIP. Relative to the position thus obtained, the lifting height can be increased by hyper extending the MCP. This can be realised by flexing the PIP further than strictly required to take up ϵ_{ED_e} , thus creating a shortage of length in the extensor tendon which drives the MCP into hyper extension. Hereby tendon length is transferred from the MCP to the PIP (Leijnse *et al.*, 1992).

For further use, the change of position of F_2 during the key stroke of F_1 will be called a '*conjoined movement*'. Conjoined movements can generally be defined as follows. Take a problem $p=(p_1, p_2, \dots)$ in which some fingers, but not all, have to solve an identity problem. Then conjoined movements in a solution $s=(s_1, s_2, \dots)$ are any changes in the positions ($d\theta_j^{(i)} \neq 0$) of fingers F_i of which the individual problems p_i are identity problems.

As is clear from the example, conjoined movements result from the fact that the connected tendons (i.e. the common motor M_e) have a non-zero displacement. This non-zero displacement requires the simultaneous coordination of all connected fingers even if the problem itself is essentially monodigital (only one finger solves a non-identity problem), as in the present example. Note that in the example the conjoined movement is necessary to obtain a solution. This illustrates that finger problems exist for which the

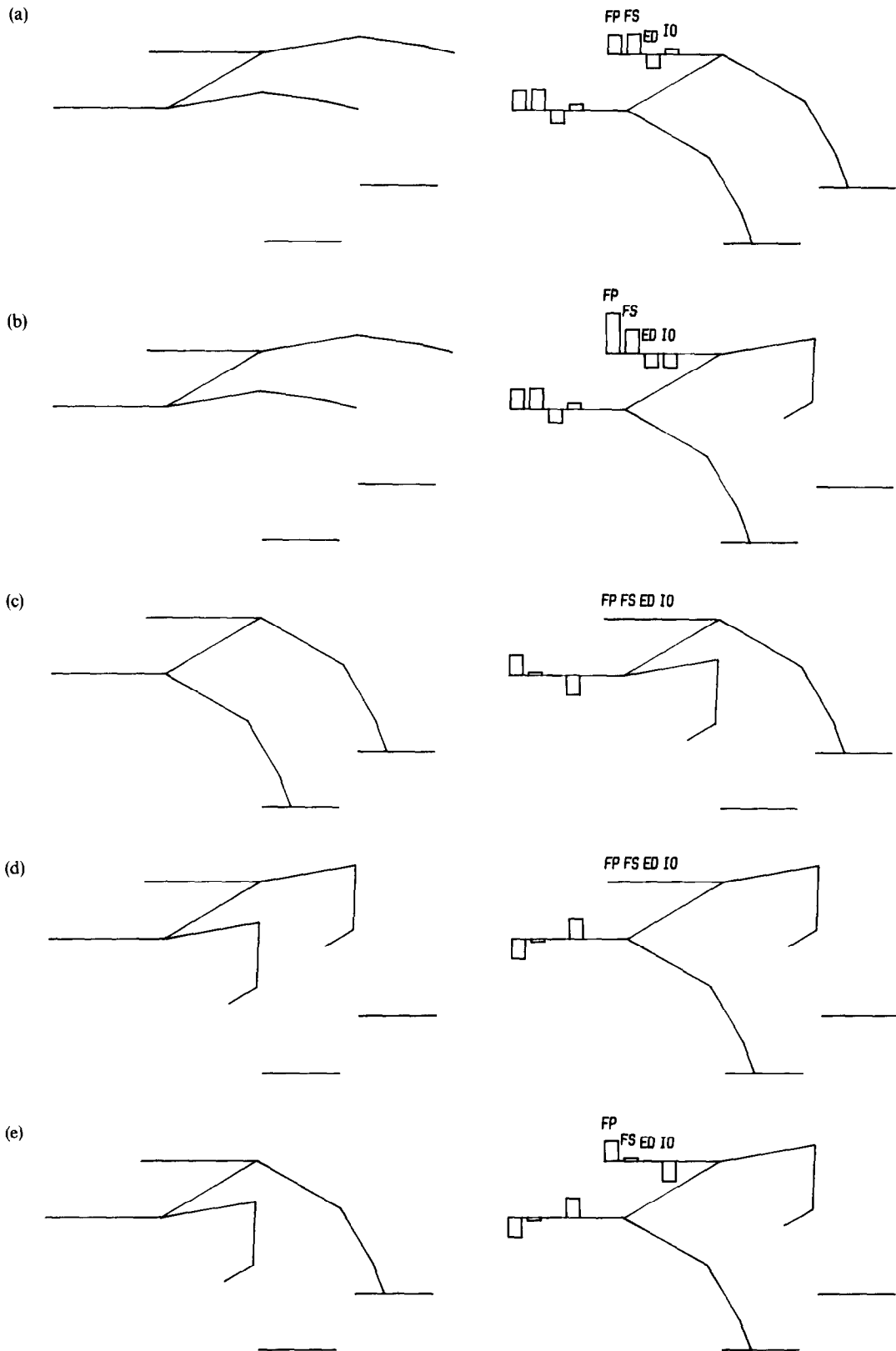


Fig. 5. Movements of the bidigital system with connected extensors: (a) Unimodal solution to the problem $[(U, U), (k, k)]$; (b) Iso-excursion solution to the problem $[(U, U), (k, l)]$; (c) Isometric solution to the problem $[(D, D), (l, k)]$; (d) Isometric solution to the problem $[(U, U), (k, l)]$; (e) Isometric solution to the inverse-symmetric problem $[(U, D), (k, l)]$.

feasible solutions in the connected system are inherently more complex than in the unconnected system, where monodigital problems can always be solved by monodigital solutions, i.e. by moving just one finger.

Problem $[(D, D), (l, k)]$ [Fig. 5(c)]. The initial position of both fingers is the end position of the striking finger in the previous solution. From this position, F_1 has to lift while F_2 must solve the identity problem of keeping its key down.

Solution. As stated above, conjoined movements occur as a result of the non-zero displacement of the connected motors (i.e. the common motor M_c). This implies, however, that the conjoined movements can be avoided by using movements in which the common motor M_c does not displace (i.e. is isometric). In Leijnse *et al.* (1992) it was shown that it is possible to lift the finger from the piano key to a sufficient height while keeping a tendon isometric. The resulting solution for present problem, i.e. the lifting of the finger with an isometric ED_c , is depicted in Fig. 5(c). For further use, such solutions will be called *isometric solutions*, bearing in mind that the isometric displacement is with the common motor.

Problem $[(U, U), (k, l)]$ [Fig. 5(d)]. The second problem of this section [Fig. 5(b)] is now considered again, but with the initial position (i_1, i_2) of both fingers equal to the collapse end position of finger F_1 in Fig. 5(b). From this position, F_1 has to strike a key, while F_2 has to remain lifted.

Solution. Any conjoined movement of F_2 can be avoided when F_1 strikes the key with an isometric ED_c . This key stroke movement exists as the kinematic inverse of the lifting movement in the previous solution [Fig. 5(c)]. The result is presented in Fig. 5(d).

Problem $[(U, D), (k, l)]$ [Fig. 5(e)]. This problem is presented to demonstrate the special nature of the isometric solution in the multidigital finger system with connected tendons. The initial positions are the end positions of the previous solution [Fig. 5(d)]. From these positions finger F_1 , initially lifted, strikes a key, while F_2 , initially on the keyboard, must lift. In other words, the individual problem of each finger is the inverse of the problem of the other finger. To indicate this basic symmetry, this problem will be further called the *inverse-symmetric problem*.

Solution. From the given initial positions, solutions are possible with a (limited) non-zero displacement ϵ_{ED_c} , positive or negative, of the common motor ED_c . Hereby the end position of the finger which lifts will always be a collapse position (see Appendix). However, of all the possible solutions, only the isometric solution (with $\epsilon_{ED_c} = 0$) reflects the basic symmetry of the problem in the sense that the fingers solve their individual problems, which are each others inverse, with solutions that are also each others exact inverse (as $e_1 = i_2$ and $e_2 = i_1$).

Application: the 'thrill' on the piano as a sequence of inverse-symmetric problems. Consider a 'thrill' on the piano. This is a sequence (or 'chain') of inverse-symmetric problems in which each problem is the inverse of the predecessor. This problem chain can be completely solved by the isometric solution for the starting problem and its exact inverse [interchange the digits in Fig. 5(e)]. Hereby the symmetry of the problem chain is reflected in the solution chain. Non-isometric solutions cannot generate solution chains with the same degree of symmetry. This is because the non-zero displacement (ϵ_{M_c}) of the common motor M_c in the solution of the starting problem will have to be inverted ($-\epsilon_{M_c}$) in the solution of the next (inverse) problem, otherwise the displacement of the M_c would keep increasing as the problem chain is executed. These symmetry/asymmetry properties have implications on the level of control. When the problem chain is executed very fast, as in a thrill on the piano, a solution chain that has the same symmetry as the problem chain should be simpler to execute than a less symmetric chain. Therefore it can be speculated that in the real hand with strictly connected tendons ($L_c = 0$), chains of inverse-symmetric problems will tend to be solved by isometric solutions.

The bidigital system with strictly connected flexor tendons

In this section, the bidigital system with strictly ($L_c = 0$) connected flexor tendons (FS, FP) is considered. Substitution of equation (1a) or (1b) into equation (9) with $M_c = F_c$ (F stands for either the FS or the FP) produces the condition imposed by the connection on the movements of the connected fingers:

$$\sum_{j=1}^{n_{F_1}} r_{F_j} d\theta_j^{(1)} - d\sigma_{F_a}^{(1)} = \sum_{j=1}^{n_{F_2}} r_{F_j} d\theta_j^{(2)} - d\sigma_{F_a}^{(2)} \quad (15)$$

The case of connected flexors is fundamentally different from the case of connected extensors, because the finger can still function properly when one of the flexors is slack, while it cannot do so with a slack extensor. When the ED is slack, the free-moving finger is uncontrollable at the MCP, for which the ED is the only extensor. However, the flexors can partly substitute each other, as expressed by condition (1f). For instance, when the FP is slack, the proximal two joints of the finger can be properly coordinated by the FS, ED and IO —be it that the position of the DIP cannot be controlled. Conversely, when the FS is slack, the finger can be properly coordinated by the FP, ED and IO . Therefore, when a flexor tendon of one finger is connected with a flexor tendon of the other, three kinds of finger movements must be considered, i.e. the movements in which (i) both connected tendons are taut, (ii) both connected tendons are slack (meaning that the common motor is inactive) and (iii) one connected tendon is taut, and one is slack.

(i) *Movements with both connected flexor tendons taut* ($\sigma_F^{(1)} \equiv \sigma_F^{(2)} \equiv 0$). When both connected flexor ten-

dons are taut, the case is similar to the ED-connected case. Only iso-excursion solutions are feasible, with the mentioned important applications for the *uni-modal*, and *isometric* solutions (no conjoined movements).

(ii) *Movements with both connected flexor tendons slack* ($\sigma_F^{(1)}, \sigma_F^{(2)} \neq 0$). When all connected flexor tendons are slack (in other words, when the common motor is inactive), the connection also will be slack and therefore inactive as a constraint. The fingers will then move unconstrained as if the connections were not present at all. In other words, when the connected flexors are not used at all, the coupling between the finger actions, as resulting from the connection, does not become manifest. Consequently, movements with inactive (slack) connected flexors will be further called *decoupling solutions*. In these solutions, however, the fingers will function with the disadvantages resulting from the non-use of the connected motors. For instance, in the case of FP tendons connected with each other, the DIP remains uncontrollable when the FP is inactive. In the FS connected case, when the FS is inactive, the PIP will lack flexion force.

(iii) *Movements with one connected flexor tendon taut and one slack*. When one flexor tendon is taut (say in F_1), the other finger (F_2) may execute unconstrained any movement as long as its connected flexor is slack ($\sigma_{T_c}^{(2)} \geq 0$). However, this finger (F_2) will hereby function with the disadvantages resulting from the non-use of this tendon $T_c^{(2)}$.

The initial positions of the fingers as determining factors of a solution

The initial positions of the fingers determine to a large extent the kind of solutions that can be used. For example, in the problem of Fig. 5(b) an isometric solution is not feasible because from the initial positions (extended fingers) the striking finger cannot reach the key except if the common motor, the ED_c, is allowed a non-zero distal displacement.

The movements of fingers with tendons connected with connections of non-zero length ($L_c > 0$)

In this section, connections of non-zero length ($L_c > 0$) are investigated, with the focus on the effect on finger function of the initial position of the constraint [parameter ΔL_0 in equations (4) and (5)]. Hereby ΔL_0 is defined with respect to a "neutral position" of the fingers, being that in which both fingers are in identical positions ($\theta_j^{(1)} = \theta_j^{(2)}$, $j = 1 \dots 3$, with all tendon parts taut. Two cases are compared. In the first case the connection is taut ($\sigma_c = 0$) in the neutral position, meaning that $\Delta L_0 = L_c$ [from equation (5) with $\sigma_c = \Delta \varepsilon_{T_c} = 0$] and in the other the connection is maximally slack ($\sigma_c = L_c$) in the neutral position, meaning that $\Delta L_0 = 0$ [from equation (5) with $\sigma_c = L_c$ and $\Delta \varepsilon_{T_c} = 0$].

(i) *Movements with the connection taut in the neutral position* ($\sigma_{c0} = 0$). When the connection is taut with the fingers in the neutral position, the functioning of

the bidigital system is asymmetric. This may be formally demonstrated as follows. When in a bidigital problem $p = (p_1, p_2)$ the digits are interchanged (to produce the mirror problem p' with $p'_1 = p_2$ and $p'_2 = p_1$), the interchanging of the digits in a random solution $s = (s_1, s_2)$ of p will not result in a feasible solution s' (with $s'_1 = s_2, s'_2 = s_1$) of p' , except when this solution is an iso-excursion solution ($\Delta \varepsilon_{T_c} = 0$). In other words, the mirror image of any solution in which the connection becomes slack cannot be applied to solve the mirror problem p' . This asymmetry is illustrated by the fact that two superimposed taut connections of opposite orientation (see paragraph 3 of "displacement properties of connected tendons" in methods and materials) allow only movements in which [from equation (4)]:

$$\left. \begin{aligned} T_{c_p}^{(1)} T_{c_d}^{(2)} &\rightarrow -2L_c \leq \Delta \varepsilon_{T_c} \leq 0 \\ T_{c_p}^{(2)} T_{c_d}^{(1)} &\rightarrow 0 \leq \Delta \varepsilon_{T_c} \leq 2L_c \end{aligned} \right\} \Rightarrow \Delta \varepsilon_{T_c} = 0 \quad (16)$$

which is the isoexcursion condition as resulting from a connection of zero length [$L_c = 0$ in equation (5)].

(ii) *Movements with the connection maximally slack in the neutral position* ($\sigma_{c0} = L_c$). In contrast, when the connection is initially maximally slack ($\Delta L_0 = 0$) with the fingers in the neutral position, a symmetrical functioning system results. Formally, when in the problem $p = (p_1, p_2)$ the digits are interchanged (problem p' with $p'_1 = p_2$ and $p'_2 = p_1$), the interchanging of the digits in any solution $s = (s_1, s_2)$ of p will result in a feasible solution s' (with $s'_1 = s_2, s'_2 = s_1$) of p' . Moreover, the corresponding phases of s and s' will be constrained in exactly the same way.

Application: the pianistic 'thrill' with $L_c \neq 0$. In the above example of the pianistic thrill the symmetry relationships between chains of inverse-symmetric problems and their solutions were investigated for connections of zero length between tendons ($L_c = 0$). In this paragraph these relationships are investigated for connections of non-zero length ($L_c \neq 0$).

(i) In the asymmetric system ($\sigma_c = 0$ in the neutral position) holds the systemic condition for symmetric functioning: $\Delta \varepsilon_{T_c} = 0$, which means that only iso-excursion solutions can be "mirrored" to solve a problem p and its mirror problem p' ($p'_1 = p_1, p'_2 = p_2$). However, under this condition the system functions identically to the system with connections of zero length [see equation (8)]. The requirement of having a symmetric solution chain for the inverse-symmetric problem chain of the pianistic thrill puts an additional condition of symmetry on these iso-excursion movements, i.e. that the end position of one finger should equal the begin position of the other and *vice versa* ($e_1 = i_2$ and $e_2 = i_1$). Within the class of the iso-excursion solutions, this condition can only be satisfied by the isometric solutions ($\varepsilon_{Mc} = 0$). It follows that under the given demands for symmetry, the bidigital system with taut connections of non-zero length in the neutral position behaves identically to the system with connections of zero length.

(ii) On the other hand, in the symmetric system with the connection maximally slack ($\sigma_c = L_c$) in the neutral position, any solution of an inverse-symmetric problem can be inverted to solve the inverse problem, because the slack range (5) of the connection is symmetric.

It follows that, in addition to its length L_c , the orientation of a connection is an important parameter in the functioning of the multifinger system with connected tendons.

Extensible connections between tendons ($dL_c \neq 0$)

When the connection is taut and extensible [$dL_c \neq 0$ in equation (7)], the relative displacements $\Delta\epsilon_{Tc}$ of the connected tendons can be increased by the use of force:

$$dF_c = k dL_c = k \cdot \text{sign} [\Delta\epsilon_{Tc} - \Delta L_0]_{\sigma_c=0} \cdot d\Delta\epsilon_{Tc} \quad (17)$$

with $\sigma_c = 0$, k the stiffness of the connection, and F_c the force by which it is stretched. In that case, movements that would be inhibited if the connection were inextensible, may be executable with effort. Clearly, the stiffer the connections are (by build, or by the fact that they become stretched to their limits), the more their functioning will be similar to the inextensible connections.

DISCUSSION

Anatomical connections within the muscles of the fingers

The results obtained in this study show that interconnections between tendons of different fingers cause limitations in the movements of these fingers. These limitations in movement are of the same nature as the limitations caused by restrictions on the displacements of the tendons of the individual finger, as studied in Leijnse *et al.* (1992). Although the present paper only models connections between tendons, other anatomical causes of restricted multidigital finger movement have been pointed out, e.g. by Kaplan (1965), whose views are discussed in this section.

Kaplan notes that the (extrinsic) motors of the fingers originate from muscle groups in which the muscle parts running to the individual tendons do not always form independent muscle bellies. Often, these muscle parts are anatomically strongly interconnected, so that the displacements of these parts relative to each other are limited. Kaplan describes an experiment in which he electrically activates a muscle part, and observes that the connected muscle parts are pulled with it. Although Kaplan does not state so explicitly, from this description two main causes of functional dependency between muscle parts to individual fingers may be outlined. The first is anatomical interconnections, as muscle parts are interconnected by connective tissues. The second may be called 'co-innervation': in the poorly individualised muscle group it cannot be excluded that muscle fibres insert-

ing in one tendon are motorically conjoinedly innervated with the muscle part of another tendon. As Kaplan observed, dependencies within the muscle will produce similar effects as the (inextensible) connections between tendons studied in this paper. When one muscle part contracts, or is pulled distally by its tendon, the connected parts must follow this displacement; if they do not, this displacement is inhibited. However, since dependencies within the muscle are to be considered extensible (muscles or connective tissues can be stretched), their effects may be to a degree amendable by the use of force, as indicated in the results [equation (17)].

Clearance of anatomical connections

The idea of surgically removing connections between tendons, in order to obtain greater finger independence, is not new (Kaplan, 1965; Forbes, 1991; McGregor and Glover 1988). The results of this paper indicate that such a procedure is in principle viable, as it would facilitate or allow the execution of finger movements otherwise difficult or inexecutable. Clinically, the clearance of well-defined connections between tendons may be feasible. However, the clearance of connections within the muscle may not be feasible. The problem then is, as Kaplan (1965) stated, that the isolated clearance of the connections between the tendons will merely leave the fingers independent up to the level of the muscle.

It is therefore important to know the exact connective structure within the muscle-tendon groups. This may be investigated by the testing of finger function. However, it hereby holds that when constraints of varying extensibility are simultaneously present, the lesser constraints will be masked by the stiffer and stricter, and can only be detected within the slack range of the latter. Therefore, it is difficult or impossible [as e.g. in the case of inextensible connections of zero length between tendons ($L_c \rightarrow 0$)] to obtain by mere function tests complete information about the connections within the musculotendinous apparatus.

Finger independence and exercise

The limited ability to execute finger movements is frequent in musicians and is known as lack of finger independence. Because it is disturbing in the playing of the instrument, exercises have been devised to overcome it. Many of these exercises have the specific purpose of increasing the net range of finger independence (i.e. the maximal amplitude of relative motion between the fingers), and attempt this by forcing the fingers, in one way or another, to exceed their normal range of independent movement. However, since finger independence is ultimately determined by the relative displacements possible between the motors, the real purpose of such exercises is to increase the relative motor displacements. Hereby it is questionable whether stiff tendinous connections or connections within the muscle can be significantly and permanently stretched by such procedures. If this is not the case, exercises of this kind are useless, the more so since

they may in their excess damage the musculotendinous apparatus.

A pedagogy tailored to the musicians hand

The above discussion suggests that anatomical variations in the hand are systemic parameters, more or less invariant to exercise. In this view, each hand has unique functional characteristics, determined by its unique anatomical built. This perception cautions against too strict a concept of the 'ideal' instrumental technique. Especially in piano playing, many attempts have been made to formulate such a codex (of the 'ideal' key stroke, the 'ideal' hand position, etc.). However, from the discussion follows that the results of these investigations cannot properly be applied without taking into account the anatomic parameters of the individual hand. Within the view of the hand as a constrained and invariant system, the purpose of exercise can only be the selection and assimilation of the most functional movements compatible with the (anatomic) constraints, and paedagogic models should be flexible enough to be of guidance at the level of the individuals' possibilities.

CONCLUSION

In this paper, the effect on the motion of a bidigital system of tendons connected with each other is exhaustively studied. It is found that connections between tendons restrict the movements of the connected fingers in typical ways, e.g. by causing conjoined movements of the fingers, as tendons of one finger pull with them tendons of the other. These characteristic phenomena may be observed in the real hand, and are known by the musician as lack of finger independence. It can therefore be concluded that anatomical connections between tendons decrease finger independence. As an important result, it is shown that the conjoined movements may be avoided in finger movements in which the connected tendons are isometric. Further, it is indicated that the clearance of stiff anatomical constraints between tendons does not guarantee that in the real hand finger independence or control will improve, because similar handicaps may be caused by other anatomical constraints, as may be present within the muscle itself. Lastly, the study questions

$$\begin{aligned} h_{\text{Fup}}^{(2)} - h_{\text{Fup}}^{(1)} &= -l_1 [\sin(\theta_1^{(2)}) \\ &= -l_2 [\sin(\theta_1^{(2)} + \theta_2^{(2)}) \\ &= -l_3 [\sin(\theta_1^{(2)} + \theta_2^{(2)} + \theta_3^{(2)}) \\ &\geq H \end{aligned}$$

whether net finger independence can be substantially increased by exercises, and motivates that pedagogic models of the instrumental technique should take into account the anatomical possibilities of the individual hand.

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APPENDIX

The solutions to the problem [(U, U), (k, l)]; connected extensors (ED), fingers initially fully extended [Fig. 5(b)]

The problem is (see Fig. A1)

$$\begin{aligned} &-\sin(\theta_1^{(1)}) \\ &-\sin(\theta_1^{(1)} + \theta_2^{(1)}) \\ &-\sin(\theta_1^{(1)} + \theta_2^{(1)} + \theta_3^{(1)}) \end{aligned} \quad (A1)$$

with the condition that

$$r_{E1} d\theta_1^{(1)} + r_{M2} d\theta_2^{(1)} = r_{E1} d\theta_1^{(2)} + r_{M2} d\theta_2^{(2)} \quad (A2)$$

with

$$\theta_k^{(i)} = \theta_{ko}^{(i)} + \Delta\theta_k^{(i)}. \quad (A3)$$

We assume all tendon parts taut [$\sigma_F = \sigma_L = 0$ in equation (1)]. Then holds that the rotation of the DIP is coupled to the rotation of the PIP. For this coupling, a linear approximation (Leijnse *et al.*, 1992) is used:

$$d\theta_3^{(i)} = (a\theta_2^{(i)} + b) d\theta_2^{(i)}. \quad (A4)$$

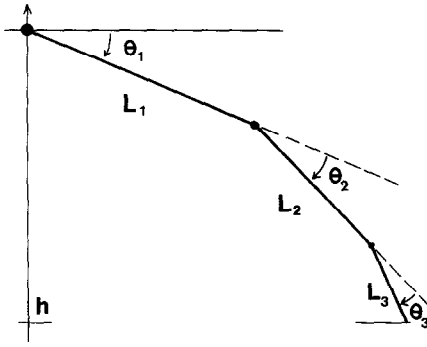


Fig. A1. Parameters of the finger model.

Integration gives

$$\Delta\theta_3^{(i)} = \frac{a}{2} [(\theta_{20}^{(i)} + \Delta\theta_2^{(i)})^2 - (\theta_{20}^{(i)})^2] + b \Delta\theta_2^{(i)}, \quad (\text{A5})$$

while integration of equation (A2) produces (all moment arms are constant):

$$\Delta\theta_2^{(2)} = \frac{r_{E1}}{r_{M2}} [\Delta\theta_1^{(1)} - \Delta\theta_1^{(2)}] + \Delta\theta_2^{(1)}. \quad (\text{A6})$$

The expressions (A5) and (A6) allow the elimination of $\Delta\theta_3^{(1)}$, $\Delta\theta_3^{(2)}$ and $\Delta\theta_2^{(2)}$ from (A1). Further holds that $\theta_{ko}^{(i)} = 0$ ($k = 1 \dots 3$), since initially all joints are fully extended. Under these conditions (A1) becomes

$$\begin{aligned} h_{\text{Ftip}}^{(2)} - h_{\text{Ftip}}^{(1)} &= -l_1 [\sin(\Delta\theta_1^{(2)}) - \sin(\Delta\theta_1^{(1)})] \\ &= -l_2 \left[\sin(\Delta\theta_1^{(2)} + \frac{r_{E1}}{r_{M2}} (\Delta\theta_1^{(1)} - \Delta\theta_1^{(2)}) + \Delta\theta_2^{(1)}) - \sin(\Delta\theta_1^{(1)} + \Delta\theta_2^{(1)}) \right] \\ &= -l_3 \left[\sin \left\{ \Delta\theta_1^{(2)} + \frac{r_{E1}}{r_{M2}} (\Delta\theta_1^{(1)} - \Delta\theta_1^{(2)}) + \Delta\theta_2^{(1)} + \frac{a}{2} \left(\frac{r_{E1}}{r_{M2}} (\Delta\theta_1^{(1)} - \Delta\theta_1^{(2)}) + \Delta\theta_2^{(1)} \right)^2 \right. \right. \\ &\quad \left. \left. + b \left(\frac{r_{E1}}{r_{M2}} (\Delta\theta_1^{(1)} - \Delta\theta_1^{(2)}) + \Delta\theta_2^{(1)} \right) \right\} - \sin \left\{ \Delta\theta_1^{(1)} + \Delta\theta_2^{(1)} + \frac{a}{2} (\Delta\theta_2^{(1)})^2 + b \Delta\theta_2^{(1)} \right\} \right]. \quad (\text{A7}) \end{aligned}$$

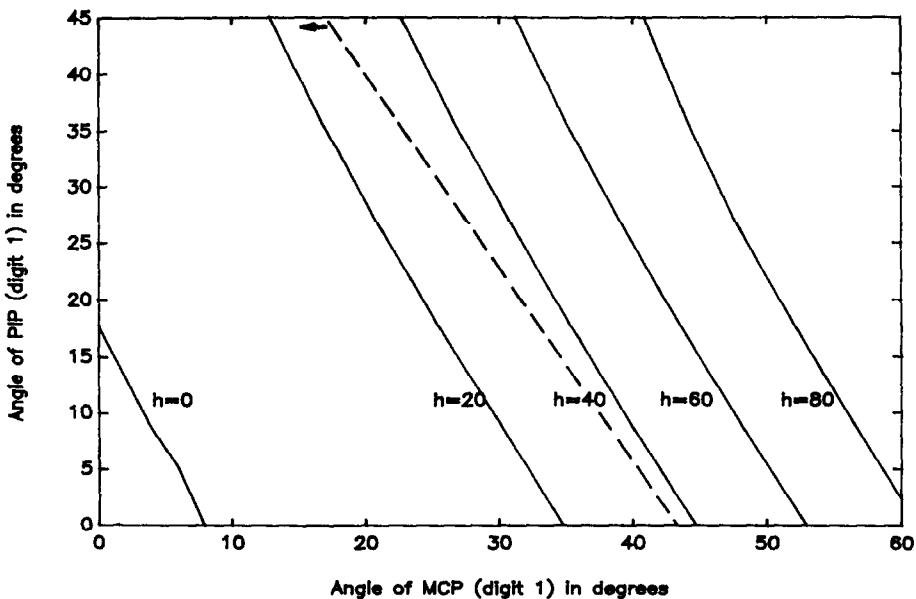


Fig. A2. Solutions to the problem: [(U, U), (k, l)], in the bidigital system with connected extensors. (—) iso-height curves of the finger tip of digit 2. (---) 100° limit of flexion of the PIP of digit 2 [the domain of feasible solutions is indicated by the little arrow (←)].

Three independent variables remain: $\theta_1^{(1)}$, $\theta_2^{(1)}$ and $\theta_1^{(2)}$. From these $\theta_1^{(2)}$ is physically determined since the lifting height of F_2 increases with the extension of the MCP⁽²⁾. This means that to achieve maximum lifting height the MCP⁽²⁾ must extend up to its functional end position, here taken to be 10° hyper extension. This puts

$$\Delta\theta_1^{(2)} = -10^\circ \quad (\text{A8})$$

and leaves two independent variables: $\theta_1^{(1)}$ and $\theta_2^{(1)}$. From the iso-height curves in Fig. A2, follow the MCP⁽¹⁾ and PIP⁽¹⁾ end positions of the candidate solutions in function of the criterium H. The domain of the feasible solutions is determined by the limit of flexion of the PIP⁽²⁾, which is taken to be 100°:

$$\Delta\theta_2^{(2)} \leq 100^\circ. \quad (\text{A9})$$

Substituting equations (A8) and (A9) into (A6) produces the dashed boundary line of the domain of feasible solutions in Fig. A2.